GENERATIVE AI ADOPTION IN HIGHER EDUCATION

Understanding University Students' Adoption of Generative AI Tools

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KEYWORDS

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Abstract Generative AI is now part of many students' academic work, but students may adopt these tools for different reasons. This study examines how perceived ease of use (PEOU), perceived usefulness (PU), and social influence (SI) relate to behavioral intention (BI) to use generative AI for academic tasks. It also tests whether PU mediates the relationship between PEOU and BI. An anonymous Qualtrics survey produced 83 recorded responses. After consent, eligibility, attention-check, and data-quality screening, 63 usable cases remained. The model was tested with partial least squares structural equation modeling. PEOU significantly increased PU (beta = .506, p < .001), and PU significantly increased BI (beta = .457, p = .002). The direct PEOU -> BI and SI -> BI paths were not significant. The indirect PEOU -> PU -> BI effect was significant (beta = .232, p = .010), indicating indirect-only mediation. The model explained 25.7% of the variance in PU and 37.3% of the variance in BI. PLSpredict also showed lower RMSE and MAE than the linear-model benchmark for all seven predicted indicators. Overall, the findings suggest that students' intention to keep using GenAI depends more on practical usefulness than on ease or social pressure alone.

1. Introduction

Generative artificial intelligence (GenAI) tools such as ChatGPT, Google Gemini, Microsoft Copilot, and Claude are becoming part of university students' academic routines. Students can use these tools to generate ideas, explain difficult concepts, organize information, revise writing, and support productivity. Yet access alone does not explain why a student plans to keep using a tool. Adoption depends on how students evaluate the technology and the value they receive from it.

Technology acceptance research provides a useful way to study this question. The Technology Acceptance Model (TAM) proposes that perceived usefulness and perceived ease of use are central beliefs that shape technology acceptance (Davis, 1989; Davis et al., 1989). Social influence is also important in broader acceptance models because people may be affected by the views of instructors, peers, friends, or other important people (Venkatesh et al., 2012). These ideas fit the GenAI context well. Students can judge whether GenAI is easy, whether it improves academic work, and whether people around them support its use.

This study applies these ideas to university students who have already used GenAI for an academic task. The model includes four reflective constructs: perceived ease of use (PEOU), perceived usefulness (PU), social influence (SI), and behavioral intention (BI). Age and prior GenAI experience are included as manifest control variables. The model also tests whether PU explains how PEOU is connected with BI.

The research question is: How do perceived ease of use, perceived usefulness, and social influence shape university students' behavioral intention to use generative AI for academic tasks, and does perceived usefulness mediate the effect of perceived ease of use on behavioral intention?

2. Theoretical Foundation and Hypotheses

2.1 Perceived Ease of Use and Perceived Usefulness

Davis (1989) defines perceived ease of use as the degree to which a person believes that using a technology will require little effort. In this study, PEOU refers to how easy students find learning, understanding, and becoming skillful with GenAI for academic tasks. A tool that is easier to use can

reduce the effort needed to obtain a useful result. For that reason, TAM predicts that ease of use can increase perceived usefulness.

H1: Perceived ease of use positively influences perceived usefulness of GenAI tools for academic tasks.

Perceived usefulness is the degree to which a person believes that a technology improves task performance (Davis, 1989). Here, PU reflects whether GenAI helps students work faster, perform better, increase productivity, and complete academic tasks. TAM places usefulness close to behavioral intention because people are more likely to continue using a system when they see clear performance value.

H2: Perceived usefulness positively influences behavioral intention to use GenAI tools for academic tasks.

Ease of use may also have a direct effect on intention. A tool that is difficult to learn or operate can discourage continued use. A tool that feels simple may lower this barrier. Based on TAM, the study therefore tests a direct path from PEOU to BI in addition to the path through PU.

H3: Perceived ease of use positively influences behavioral intention to use GenAI tools for academic tasks.

Figure 1. Proposed Structural Model

2.2 Social Influence

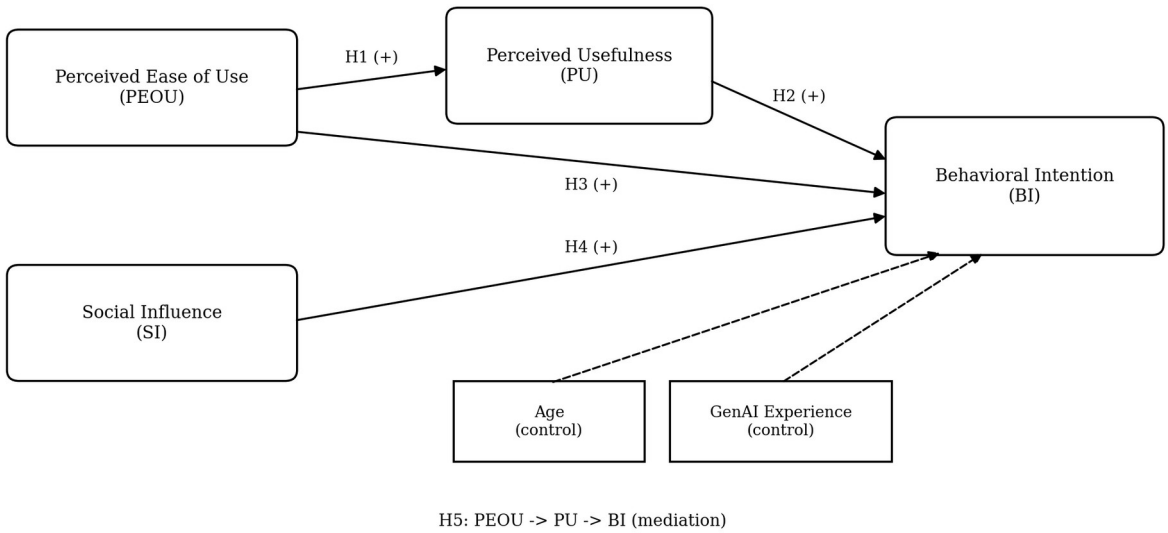
Social influence refers to the extent to which a person believes that important others think the technology should be used. This construct is central to UTAUT and UTAUT2 (Venkatesh et al., 2012). In a university setting, students may observe classmates using GenAI, receive advice from instructors, or consider the views of people who influence their academic behavior. Positive social signals may therefore strengthen intention.

H4: Social influence positively influences behavioral intention to use GenAI tools for academic tasks.

2.3 Mediating Role of Perceived Usefulness

The model also treats usefulness as a mechanism that can connect ease of use with intention. A GenAI tool can be easy to operate but still offer little academic value. Ease should matter more when it helps students obtain useful outcomes. The proposed mediation follows this logic: PEOU increases PU, and PU then increases BI.

H5: Perceived usefulness mediates the positive relationship between perceived ease of use and behavioral intention.



Note. Dashed paths represent the two manifest control variables. The mediation hypothesis is PEOU -> PU -> BI.

3. Research Methodology

3.1 Measurement Development

The survey used four reflective latent constructs. PEOU and PU were adapted from Davis (1989). BI was adapted from TAM-based intention measures (Davis et al., 1989). SI was adapted from UTAUT2 (Venkatesh et al., 2012). The item wording was adjusted so that every statement referred directly to GenAI use for academic tasks.

All latent construct items used a seven-point agreement scale from 1 = Strongly disagree to 7 = Strongly agree. PEOU and PU each used four indicators. SI and BI each used three indicators. Age was recorded as a numeric value in years. Prior GenAI experience was recorded in ordered categories and coded from 1 to 5. Age and experience were included as control variables predicting BI. Academic level, use frequency, and preferred GenAI tool were used for descriptive reporting only.

3.2 Survey Design and Data Collection

The survey was built in Qualtrics. The opening page explained the academic purpose of the study and stated that participation was voluntary. Respondents then answered two consent questions and an eligibility question. Only students who reported prior use of a GenAI tool for an academic task continued to the study measures. The survey then presented the PEOU, PU, SI, and BI items, an attention check, and background questions.

The anonymous survey was distributed online to university students through a nonprobability convenience approach. This was practical for a time-limited course project. It also means that the final sample should not be treated as a representative probability sample of all university students. The anonymous Qualtrics survey link is identified in Appendix A and should be replaced with the final live link before submission.

3.3 Sample and Data Screening

The Qualtrics export contained 83 recorded responses. Three preview or test records were removed. One respondent declined consent. Nine respondents were ineligible because they had not used GenAI for an academic task. Seven respondents failed the attention check. The final analytic sample therefore contained 63 respondents.

The retained model variables had no missing values. Completion time was inspected as a quality diagnostic. The median duration was 216 seconds. One third of the median, 72 seconds, was used only as a flag and not as an automatic deletion rule. The shortest retained response was 74 seconds, so no case was removed as a speeder. No exact duplicate substantive response patterns were found. Four retained respondents used the same answer across all 14 positively worded construct items. They passed the attention check, so there was not enough evidence to remove them.

The item distributions were also inspected. Several indicators were negatively skewed because respondents tended to select the higher end of the agreement scale. PU1 had skewness near -1.86 and excess kurtosis near 5.46, while PU3 had skewness near -2.06 and excess kurtosis near 8.13 in the SmartPLS distribution output. The Likert indicators were not transformed. PLS-SEM is designed to work without the strict multivariate normality assumptions required by covariance-based SEM, although strong nonnormality remains relevant when interpreting bootstrap uncertainty (Hair et al., 2022).

Table 1. Characteristics of the Final Study Sample (N = 63)

Characteristic	Category / statistic	n	%
Age (years)	Mean = 32.90; SD = 7.42; range = 21-54		
Academic level	Undergraduate	10	15.9%
	Master’s	25	39.7%
	Doctoral	21	33.3%
	Other	7	11.1%
GenAI experience	1–6 months	4	6.3%
	7–12 months	7	11.1%

Characteristic	Category / statistic	n	%
	1–2 years	29	46.0%
	More than 2 years	23	36.5%
Use frequency	Rarely	1	1.6%
	Sometimes	15	23.8%
	Often	13	20.6%
	Very Often	21	33.3%
	Daily or almost daily	13	20.6%
Preferred GenAI tool	ChatGPT	41	65.1%
	Google Gemini	7	11.1%
	Microsoft Copilot	4	6.3%
	Claude	9	14.3%
	Other	2	3.2%

Note. Percentages are based on the final usable sample and may not total exactly 100% because of rounding.

3.4 PLS-SEM Analysis Procedure

The model was analyzed with partial least squares structural equation modeling (PLS-SEM). This method was suitable because the study focused on explanation and prediction, included latent constructs measured with multiple indicators, and used a small-to-medium sample. The standard PLS algorithm used path weighting and standardized results. Bootstrapping used 5,000 resamples, two-tailed tests, and a 5% significance level.

The largest number of arrows entering an endogenous construct was five for BI: PEOU, PU, SI, age, and experience. Using the 10-times guideline discussed in class, the rough minimum sample was therefore 50. The final N of 63 exceeded this heuristic. The rule is only a rough guideline and is not a substitute for formal power analysis.

The analysis followed two stages. The reflective measurement model was assessed first. Outer loadings, Cronbach's alpha, rho_A, composite reliability, AVE, HTMT, Fornell-Larcker, and cross-loadings were examined. The structural model was then assessed using inner VIF, path coefficients, bootstrap confidence intervals, R2, adjusted R2, f2, mediation, and out-of-sample prediction. PLSpredict used 10 folds and 10 repetitions in R because the available SmartPLS installation did not expose the PLSpredict procedure. SmartPLS values remained the primary source for all analyses that were available in SmartPLS.

4. Results

4.1 Descriptive Results

The final sample had a mean age of 32.90 years (SD = 7.42), with a range from 21 to 54. Master's students were the largest academic-level group (39.7%), followed by doctoral students (33.3%), undergraduates (15.9%), and students who selected Other (11.1%). Most respondents had substantial prior GenAI experience. Forty-six percent reported 1-2 years and 36.5% reported more than two years. ChatGPT was the preferred tool for 65.1% of respondents.

The construct means were high for PEOU (M = 5.90, SD = 0.83), PU (M = 5.80, SD = 0.93), and BI (M = 5.96, SD = 1.00). SI was lower and more dispersed (M = 4.64, SD = 1.30). This pattern means that respondents generally viewed GenAI as easy and useful and reported strong intention to keep using it. Social influence was less consistent across students.

4.2 Reflective Measurement Model

The measurement model was evaluated before any structural path was interpreted. Most outer loadings exceeded the preferred level of about .708. PEOU2, PEOU3, and PEOU4 loaded at .850, .842, and .810. PU loadings ranged from .778 to .861. SI loadings ranged from .918 to .940. BI loadings ranged from .860 to .968. These values

show that the indicators generally represented their intended constructs well.

PEOU1 was the only clearly lower indicator, with a loading of .630. It was not removed automatically. The PEOU construct still showed satisfactory reliability and AVE, and the item represented the learning-ease part of the construct. Removing it only to increase the loading profile would have reduced content coverage. The item was therefore retained.

Table 2. Reflective Measurement Model Assessment

Construct	Indicator	Loading	Cronbach alpha	rho_A	CR	AVE
PEOU	PEOU1	.630	.798	.842	.866	.621
	PEOU2	.850				
	PEOU3	.842				
	PEOU4	.810				
PU	PU1	.781	.829	.851	.885	.659
	PU2	.861				
	PU3	.778				
	PU4	.825				
SI	SI1	.933	.923	.949	.951	.865
	SI2	.918				
	SI3	.940				
BI	BI1	.968	.914	.960	.945	.852
	BI2	.937				
	BI3	.860				

Note. CR = composite reliability; AVE = average variance extracted. PEOU1 was retained because PEOU reliability and AVE remained satisfactory and the item added theoretical content.

Internal consistency was satisfactory for every substantive construct. Cronbach's alpha was .798 for PEOU, .829 for PU, .923 for SI, and .914 for BI. rho_A was .842, .851, .949, and .960, respectively. Composite reliability was .866 for PEOU, .885 for PU, .951 for SI, and .945 for BI. The values for PEOU and PU sit comfortably in the usual reliability range. SI and BI were very high. SI's CR of .951 reaches the level where item similarity should be considered. The items were still retained because they refer to different forms of social influence and because the scale contained only three items.

Convergent validity was also supported by the AVE point estimates. AVE was .621 for PEOU, .659 for PU, .865 for SI, and .852 for BI. All four exceeded .50, which means that each construct explained more than half of the variance in its indicators on average. The bootstrap bias-corrected AVE interval for PEOU was wider and extended below .50 at the lower end [.461, .740]. This does not change the point-estimate decision, but it shows that PEOU was the least stable of the four measurement blocks and supports the decision to discuss PEOU1 openly rather than ignore it.

Discriminant validity was assessed primarily with HTMT. For the substantive constructs, HTMT values were .585 for PEOU-PU, .279 for PEOU-SI, .328 for PEOU-BI, .468 for PU-SI, .615 for PU-BI, and .368 for SI-BI. Every value was below the conservative .85 criterion. The largest value was PU-BI at .615, and its 95% bootstrap interval [.414, .798] also remained below .85. PEOU-PU had the next largest HTMT at .585 with an upper confidence limit of .834. These results provide strong evidence that the four constructs were empirically distinct.

Table 3. HTMT Discriminant Validity Matrix

Construct	PEOU	PU	SI	BI
PEOU	-	.585	.279	.328

Construct	PEOU	PU	SI	BI
PU		-	.468	.615
SI			-	.368
BI				-

Note. All substantive HTMT values were below .85. Bootstrap intervals for the largest pairs also remained below .85.

The Fornell-Larcker criterion gave the same conclusion. The square roots of AVE were .788 for PEOU, .812 for PU, .930 for SI, and .923 for BI. Each diagonal value exceeded the construct's correlations with the other substantive constructs. The cross-loading check also passed. Every indicator loaded highest on its assigned construct. For example, BI1 loaded .968 on BI while its largest cross-loading was .599 on PU. PU4 loaded .825 on PU while its largest cross-loading was .582 on BI. The complete Fornell-Larcker and cross-loading tables are included in the appendices.

Outer VIF values were not used to delete reflective indicators. Outer VIF is mainly a formative-measurement criterion. Some reflective items, especially BI1 and BI2, had high outer VIF values because they were strongly correlated, but that is not by itself a reason to remove a reflective item. The reflective assessment therefore remained based on loading, reliability, convergent validity, discriminant validity, and theoretical coverage.

4.3 Structural Collinearity and Explanatory Power

Before testing the hypotheses, structural collinearity was checked for each predictor set. All inner VIF values were below 3. Age -> BI had VIF = 1.286. Experience -> BI had VIF = 1.435. PEOU -> BI had VIF = 1.606. PU -> BI had VIF = 1.631. SI -> BI had VIF = 1.378. PEOU -> PU had VIF = 1.000 because PU had only one predictor. The bootstrap upper confidence limits were also below 3; the largest was 2.758 for PEOU in the BI equation. The path estimates were therefore not distorted by problematic predictor collinearity.

The model explained 25.7% of the variance in PU (R2 = .257; adjusted R2 = .244). PEOU alone therefore accounted for about one quarter of the observed differences in perceived usefulness. The 95% percentile bootstrap interval for PU R2 was [.086, .510]. The model explained 37.3% of the variance in BI (R2 = .373; adjusted R2 = .318). Its 95% percentile bootstrap interval was [.265, .625]. Using the course benchmarks, PU is around the weak threshold and BI falls between weak and moderate. The values are still meaningful for a compact behavioral model with only three substantive predictors and two controls.

Table 4. Structural Diagnostics and Effect Sizes

Outcome	R2	Adj. R2	Predictor	VIF	f2	f2 interpretation
PU	.257	.244	PEOU	1.000	.345	Medium; near large
BI	.373	.318	PU	1.631	.205	Medium
BI			PEOU	1.606	.002	Negligible
BI			SI	1.378	.007	Negligible
BI			Age	1.286	.043	Small
BI			Experience	1.435	.017	Negligible

Note. Approximate f2 benchmarks: .02 = small, .15 = medium, .35 = large. All structural VIF values were below 3.

The f2 values show which predictors made a meaningful contribution after the other predictors were considered. PEOU had f2 = .345 for PU. This is a medium effect that is only .005 below the .35 large-effect benchmark. PU had f2 = .205 for BI, a clear medium effect. Age had a small effect size of .043, but its path was not statistically significant. Experience had f2 = .017, PEOU had f2 = .002 for BI, and SI had f2 = .007. These three values are negligible. The effect-size pattern already suggests that the important chain in the model runs from PEOU to PU and then from PU to BI.

4.4 Structural Paths and Hypothesis Tests

H1 was supported. PEOU had a positive effect on PU, beta = .506. The bootstrap standard

deviation was .112, t = 4.538, and p < .001. The 95% percentile confidence interval [.293, .714] excluded zero. The coefficient is also substantively meaningful. A one-standard-deviation increase in PEOU was associated with about a .51-standard-deviation increase in PU.

H2 was also supported. PU positively predicted BI, beta = .457, bootstrap SD = .149, t = 3.072, p = .002. The 95% interval [.151, .741] excluded zero. This was the strongest significant direct predictor of BI. Together with f2 = .205, the result shows that usefulness had both statistical significance and practical importance in the model.

H3 was not supported. The direct PEOU -> BI path was very small, beta = .048, t = .342, p = .732, with a 95% interval [-.183, .396]. The interval crosses zero by a wide margin and f2 was only .002. Ease of use therefore did not make a meaningful direct contribution to BI after usefulness and the other predictors were included.

H4 was not supported. The SI -> BI coefficient was beta = .077, t = .580, p = .562. Its 95% interval [-.196, .315] crossed zero and f2 was .007. In this sample, social encouragement or expectations did not explain additional intention once the other predictors were considered.

The two controls were also nonsignificant. Age -> BI was beta = .186, t = 1.769, p = .077, 95% CI [-.021, .397]. It had a small f2 of .043, so the direction was positive but the evidence was not strong enough for a .05 significance decision. Experience -> BI was beta = .125, t = .914, p = .361, 95% CI [-.163, .380], with f2 = .017. Prior experience category therefore did not add a reliable direct effect on BI in this sample.

Table 5. Structural Model and Hypothesis Testing

Hyp.	Relationship	beta	Boot. mean	Boot. SD	t	p	95% CI	f2	Decision
H1	PEOU -> PU	.506	.524	.112	4.538	< .001	[.293, .714]	.345	Supported
H2	PU -> BI	.457	.466	.149	3.072	.002	[.151, .741]	.205	Supported
H3	PEOU -> BI	.048	.072	.141	.342	.732	[-.183, .396]	.002	Not supported
H4	SI -> BI	.077	.066	.133	.580	.562	[-.196, .315]	.007	Not supported
Control	Age -> BI	.186	.187	.105	1.769	.077	[-.021, .397]	.043	Not significant
Control	Experience -> BI	.125	.115	.137	.914	.361	[-.163, .380]	.017	Not significant

Note. Bootstrap inference used 5,000 resamples and two-tailed tests. The reported confidence intervals are percentile intervals.

4.5 Mediation and Total Effect

H5 was tested with the specific indirect effect. The indirect PEOU -> PU -> BI effect was beta = .232. The bootstrap mean was .241, the bootstrap SD was .090, t = 2.584, and p = .010. The percentile 95% interval was [.080, .435], and the bias-corrected 95% interval was [.073, .427]. Both intervals excluded zero. The indirect effect was therefore statistically significant.

The direct PEOU -> BI effect remained nonsignificant at beta = .048 and p = .732. According to the mediation decision logic used in class, a significant indirect effect combined with a nonsignificant direct effect is indirect-only mediation. This is often described as full mediation in practical reporting. H5 was therefore supported.

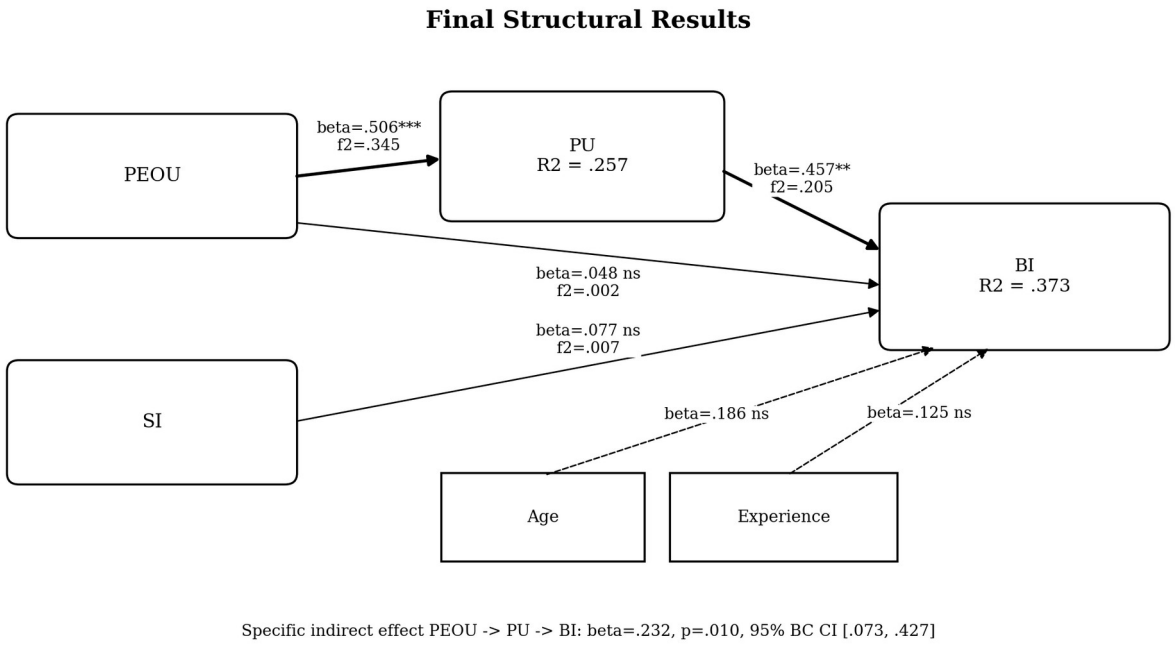
The total PEOU effect on BI can be expressed as the direct effect plus the specific indirect effect: .048 + .232 = approximately .280. This total coefficient shows that PEOU still had an overall positive association with BI, but most of that association operated through usefulness rather than through a separate direct path. This is the central mechanism found in the study.

Table 6. Mediation and Total-Effect Summary

Hyp.	Indirect path	Indirect beta	Boot. mean	Boot. SD	t	p	95% pct. CI	95% BC CI	Direct beta	Total beta*	Type
H5	PEOU -> PU -> BI	.232	.241	.090	2.584	.010	[.080, .435]	[.073, .427]	.048	.280	Indirect-only

*Note. *Total beta is computed as direct beta + specific indirect beta. BC = bias-corrected. The total-effect significance was not separately inferred from this arithmetic calculation.*

Figure 2. Final Structural Results



*Note. *** p < .001; ** p < .01; ns = not significant. Effect-size values are shown beside the two theoretically important direct paths.*

4.6 Out-of-Sample Predictive Performance

PLSpredict was used to evaluate whether the model could predict held-out observations rather than only explain the same data used to estimate it. The procedure used 10-fold cross-validation with 10 repetitions. Prediction errors from the PLS model were compared with a linear-model (LM) benchmark. Lower RMSE or MAE indicates better prediction.

The PLS model produced lower RMSE and lower MAE than the LM benchmark for every one of the seven endogenous indicators. For PU1, PLS RMSE was 1.079 compared with 1.241 for LM. For PU2, the values were 1.238 versus 1.377. For PU3, they were .965 versus 1.046, and for PU4, .979 versus 1.028. The same pattern held for BI1 (.941 vs .974), BI2 (1.165 vs 1.230), and BI3 (.869 vs 1.037). Under the course comparison rule, all seven indicators outperforming the LM benchmark indicates high predictive power relative to that benchmark.

MAE gave the same conclusion. PLS MAE was lower than LM MAE for all seven indicators. This is useful because RMSE places more weight on large errors, whereas MAE is less sensitive to extreme prediction errors. Agreement between the two metrics gives stronger evidence that the PLS model did not outperform LM only because of one error metric.

Q2predict was positive for six of the seven indicators. Positive values were observed for PU1 (.122), PU2 (.067), PU3 (.057), PU4 (.203), BI1 (.215), and BI2 (.175). BI3 had Q2predict = -.027. This means that BI3 did not beat the simple mean-based benchmark on the Q2predict criterion. The negative value was small, and BI3 still had clearly lower RMSE and MAE than the LM benchmark. The combined interpretation is therefore strong overall prediction with one weaker BI indicator on the mean-benchmark criterion, not perfect prediction for every item.

Table 7. PLSpredict Results

Indicator	PLS RMSE	LM RMSE	PLS MAE	LM MAE	Q2predict	PLS vs. LM
PU1	1.079	1.241	.745	.845	.122	PLS better
PU2	1.238	1.377	.921	1.003	.067	PLS better
PU3	.965	1.046	.642	.698	.057	PLS better

Indicator	PLS RMSE	LM RMSE	PLS MAE	LM MAE	Q2predict	PLS vs. LM
PU4	.979	1.028	.682	.731	.203	PLS better
BI1	.941	.974	.718	.744	.215	PLS better
BI2	1.165	1.230	.866	.947	.175	PLS better
BI3	.869	1.037	.682	.820	-.027	PLS better

Note. PLS had lower RMSE and MAE for all seven indicators. Q2predict was positive for six indicators and slightly negative for BI3.

4.7 Supplemental Model Fit

Model-fit indices were treated as supplemental because PLS-SEM is prediction oriented. SRMR was .092 for the saturated model and .119 for the estimated model. The saturated value was slightly above its 99% bootstrap reference value of .091. The estimated SRMR was above the 95% reference value of .112 but below the 99% value of .138. The d_ULS pattern was also mixed: the saturated value of 1.145 was slightly above its 99% reference value of 1.129, while the estimated value of 1.941 was above the 95% value of 1.708 but below the 99% value of 2.607. By contrast, d_G was within the 95% bootstrap reference values for both the saturated and estimated models. These results do not show uniformly strong approximate fit.

The mixed fit evidence is reported rather than hidden. It supports a cautious interpretation of the model. The core conclusions therefore rely on the validated measurement model, low structural VIF, significant paths, R2, f2, mediation, and out-of-sample prediction. The model should not be presented as a perfect representation of all factors that drive GenAI adoption.

Table 8. Supplemental Approximate Model-Fit Results

Index	Model	Observed	Bootstrap mean	95%	99%	Interpretation
SRMR	Saturated	.092	.066	.082	.091	Observed slightly above 99% reference
SRMR	Estimated	.119	.082	.112	.138	Between 95% and 99% references
d_ULS	Saturated	1.145	.606	.917	1.129	Observed slightly above 99% reference
d_ULS	Estimated	1.941	.953	1.708	2.607	Between 95% and 99% references
d_G	Saturated	.604	.626	.969	1.180	Within 95% reference
d_G	Estimated	.667	n/a	1.101	1.632	Within 95% reference

Note. Fit indices are supplemental in this prediction-oriented PLS-SEM study. The evidence is mixed rather than uniformly strong.

5. Discussion

5.1 Main Findings

The strongest result is the role of perceived usefulness. PU was the main significant direct predictor of BI. Students were more likely to intend to keep using GenAI when they believed that it helped them complete academic work faster, improved performance, and increased productivity. This finding is consistent with TAM, which treats usefulness as a central driver of acceptance (Davis, 1989; Davis et al., 1989).

PEOU strongly predicted PU but did not directly predict BI. The mediation result explains this pattern. Ease mattered because it helped students see GenAI as useful. Once PU was included, there was almost no direct PEOU contribution left. This means that an easy interface alone may not be enough to create continued academic use. Students appear to connect ease with intention when ease produces useful academic outcomes.

The SI result was different from the expected UTAUT2 pattern. Social influence was not a significant direct predictor of BI. One interpretation is that academic GenAI use may be more utility driven than socially driven for students who have already tried the technology. They may continue using it because it works for them, not because other people tell them to use it. This explanation should remain cautious because the sample was small and the SI measure combined several types of important others rather than separating instructors, classmates, friends, and family.

The controls also did not materially change the story. Age had a positive coefficient and a small effect size, but p = .077 did not meet the .05 criterion. Prior GenAI experience was clearly nonsignificant. In this sample, the belief that GenAI is useful was more important than simple age or experience differences.

The explanatory and predictive results tell a consistent but not perfect story. BI R2 of .373 means that the model explained a meaningful portion of intention, but most variance remained unexplained. At the same time, PLS prediction errors were lower than the LM benchmark for every endogenous indicator. The model therefore showed useful predictive ability even though it was deliberately compact. The single negative Q2predict for BI3 and the mixed supplemental fit indices show that there is still room to improve the model.

5.2 Practical Implications

Universities should focus on useful academic outcomes rather than simply promoting GenAI as a new technology. Training can demonstrate concrete tasks such as brainstorming, explanation, planning, revision, and productivity support. Students should also be shown situations where GenAI adds little value or where independent work is required. This keeps usefulness connected to responsible use.

Ease of use still matters because it helps create usefulness. Orientation can reduce technical friction and teach students how to ask clear questions, evaluate outputs, and refine prompts. However, training should not stop at interface features. It should show how an easy interaction becomes a better academic workflow.

The weak SI path suggests that simple peer pressure or general encouragement may not create lasting adoption. Universities may get better results from practical demonstrations, clear academic-integrity guidance, and faculty examples that show appropriate use. The goal should be informed, purposeful use rather than use for its own sake.

5.3 Limitations and Possible Improvements

First, the final sample was small and based on convenience sampling. N = 63 exceeded the class 10-times heuristic, but the sample does not represent all university students. A larger multi-university sample would improve precision and generalizability.

Second, the study was cross-sectional and self-reported. The paths describe statistical relationships among perceptions measured at one point in time. They do not prove strong causal effects. A longitudinal design could measure PEOU and PU first and then observe later GenAI use.

Third, only students with prior academic GenAI use were eligible. The model therefore explains continued or future intention among existing users. It does not directly explain first-time adoption among students who have never used GenAI for coursework.

Fourth, all substantive indicators were positively worded and collected in one survey. This design can increase common-method concerns. Future research could combine survey measures with actual usage logs, course performance, instructor ratings, or time-separated measurement.

Fifth, the SI scale was highly reliable and may contain some redundancy. Future work could separate instructor influence, peer influence, institutional norms, and family or friend influence. This may show whether one social source matters even when a broad SI construct does not.

Finally, the supplemental SRMR and d_ULS results were weaker than ideal. Future research could test additional theoretically justified predictors such as trust, perceived risk, AI self-efficacy, perceived accuracy, academic-integrity concerns, habit, facilitating conditions, or task-technology fit. Model expansion should be theory driven rather than performed only to improve fit statistics.

6. Conclusion

This study tested a four-construct PLS-SEM model of university students' GenAI adoption for academic tasks. PEOU significantly increased PU, and PU significantly increased BI. The direct PEOU -> BI and SI -> BI relationships were not significant. Age and prior experience were also nonsignificant controls.

The main finding is the mediation process. The indirect PEOU -> PU -> BI effect was significant, while the direct PEOU -> BI effect was not. Ease of use therefore mattered mainly because it

increased usefulness. For universities, the practical message is simple: students are more likely to keep using GenAI when they can see clear academic value from it.

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Appendix A. Qualtrics Survey Link

Qualtrics anonymous survey link: [INSERT FINAL ANONYMOUS QUALTRICS LINK BEFORE SUBMISSION]

The exact anonymous survey URL was not preserved in the analysis files. Replace the bracketed text with the live anonymous Qualtrics link, not a preview link, before submitting the PDF.

Appendix B. Survey Codebook

Table B1. Measurement Codebook

Variable	Item / description	Scale / anchors	Source
PEOU1	Learning to use generative AI tools for academic tasks is easy for me.	1-7 agreement	Davis (1989), adapted
PEOU2	My interaction with generative AI tools for academic tasks is clear and understandable.	1-7 agreement	Davis (1989), adapted
PEOU3	It is easy for me to become skillful at using generative AI tools for academic tasks.	1-7 agreement	Davis (1989), adapted
PEOU4	Overall, I find generative AI tools easy to use for academic tasks.	1-7 agreement	Davis (1989), adapted
PU1	Using generative AI tools helps me complete academic tasks more quickly.	1-7 agreement	Davis (1989), adapted
PU2	Using generative AI tools improves my performance on academic tasks.	1-7 agreement	Davis (1989), adapted
PU3	Using generative AI tools increases my productivity in academic work.	1-7 agreement	Davis (1989), adapted
PU4	Overall, I find generative AI tools useful for completing academic tasks.	1-7 agreement	Davis (1989), adapted
SI1	People who are important to me think that I should use generative AI tools for academic tasks.	1-7 agreement	Venkatesh et al. (2012), adapted
SI2	People who influence my academic behavior think that I should use generative AI tools for academic tasks.	1-7 agreement	Venkatesh et al. (2012), adapted
SI3	People whose opinions I value support my use of generative AI tools for academic tasks.	1-7 agreement	Venkatesh et al. (2012), adapted
BI1	I intend to continue using generative AI tools for academic tasks in the future.	1-7 agreement	Davis et al. (1989), adapted
BI2	I plan to use generative AI tools regularly for academic tasks.	1-7 agreement	Davis et al. (1989), adapted
BI3	I expect to use generative AI tools for academic tasks whenever they are appropriate and useful.	1-7 agreement	Davis et al. (1989), adapted
AGE	Age in years.	Numeric whole number	Control variable
EXP	Prior experience using GenAI tools.	1 = <1 month; 2 = 1-6 months; 3 = 7-12 months; 4 = 1-2 years; 5 = >2 years	Control variable

Note. All PEOU, PU, SI, and BI items used 1 = Strongly disagree and 7 = Strongly agree.

Appendix C. Additional Discriminant Validity Evidence

Table C1. Fornell-Larcker Criterion

Construct	PEOU	PU	SI	BI
PEOU	.788			

Construct	PEOU	PU	SI	BI
PU	.506	.812		
SI	.266	.422	.930	
BI	.305	.569	.361	.923

Note. Diagonal values are the square roots of AVE. Each diagonal value exceeded the corresponding interconstruct correlations.

Table C2. Indicator Cross-Loadings

Indicator	PEOU	PU	SI	BI
PEOU1	.630	.233	.083	.150
PEOU2	.850	.457	.203	.363
PEOU3	.842	.400	.149	.134
PEOU4	.810	.448	.346	.251
PU1	.412	.781	.135	.317
PU2	.348	.861	.398	.396
PU3	.328	.778	.462	.488
PU4	.516	.825	.351	.582
SI1	.266	.355	.933	.348
SI2	.174	.359	.918	.261
SI3	.282	.450	.940	.379
BI1	.308	.599	.348	.968
BI2	.298	.542	.422	.937
BI3	.222	.399	.187	.860

Note. Each indicator loaded most strongly on its intended construct. Values are rounded to three decimals.

Table C3. HTMT Bootstrap Intervals

Construct pair	HTMT	95% percentile CI
PEOU - PU	.585	[.316, .834]
PEOU - SI	.279	[.122, .537]
PEOU - BI	.328	[.155, .661]
PU - SI	.468	[.218, .707]
PU - BI	.615	[.414, .798]
SI - BI	.368	[.132, .575]

Note. All substantive HTMT point estimates and upper confidence limits remained below .85.

Appendix D. Detailed Structural Diagnostics

Table D1. Inner VIF Bootstrap Results

Structural path	Original VIF	Bootstrap mean	95% percentile CI
Age -> BI	1.286	1.374	[1.112, 1.812]
Experience -> BI	1.435	1.547	[1.192, 2.120]
PEOU -> BI	1.606	1.810	[1.229, 2.758]
PEOU -> PU	1.000	1.000	[1.000, 1.000]
PU -> BI	1.631	1.834	[1.333, 2.636]
SI -> BI	1.378	1.496	[1.163, 2.083]

Note. All VIF point estimates and 95% upper confidence limits were below 3.

Table D2. R2 Bootstrap Results

Outcome	R2	Adj. R2	Boot. mean R2	Boot. SD	95% pct. CI	95% BC CI
BI	.373	.318	.439	.092	[.265, .625]	[.167, .499]

Outcome	R2	Adj. R2	Boot. mean R2	Boot. SD	95% pct. CI	95% BC CI
PU	.257	.244	.287	.111	[.086, .510]	[.059, .464]

Note. The confidence intervals refer to R2. Adjusted R2 values are reported as point estimates.

Table D3. f2 Effect Sizes and Percentile Intervals

Structural path	f2	95% percentile CI	Interpretation
Age -> BI	.043	[.000, .236]	Small
Experience -> BI	.017	[.000, .198]	Negligible
PEOU -> BI	.002	[.000, .136]	Negligible
PEOU -> PU	.345	[.094, 1.041]	Medium; near large
PU -> BI	.205	[.020, .711]	Medium
SI -> BI	.007	[.000, .139]	Negligible

Note. Effect-size intervals are broad because the sample is small. Point estimates are used for the standard .02/.15/.35 classification.

Table D4. Model-Fit Bootstrap Reference Values

Index	Model	Observed	Mean	95%	99%
SRMR	Saturated	.092	.066	.082	.091
SRMR	Estimated	.119	.082	.112	.138
d_ULS	Saturated	1.145	.606	.917	1.129
d_ULS	Estimated	1.941	.953	1.708	2.607
d_G	Saturated	.604	.626	.969	1.180
d_G	Estimated	.667	n/a	1.101	1.632

Note. These indices are supplemental and show mixed approximate-fit evidence.

Appendix E. Item Distribution Diagnostics

Table E1. Indicator Descriptive Statistics

Indicator	Mean	SD	Skewness	Excess kurtosis	Min	Max
PEOU1	5.794	1.272	-1.395	2.353	1	7
PEOU2	6.000	0.933	-0.861	0.644	3	7
PEOU3	5.825	0.959	-0.429	-0.706	4	7
PEOU4	5.968	1.077	-1.135	1.611	2	7
PU1	5.905	1.160	-1.858	5.456	1	7
PU2	5.556	1.292	-1.145	1.072	2	7
PU3	5.889	1.002	-2.057	8.126	1	7
PU4	5.857	1.105	-1.486	2.555	2	7
SI1	4.714	1.430	-0.602	-0.161	1	7
SI2	4.444	1.446	-0.365	-0.287	1	7
SI3	4.762	1.316	-0.901	0.783	1	7
BI1	5.984	1.070	-1.191	1.151	3	7
BI2	5.683	1.293	-1.227	1.658	1	7
BI3	6.206	0.864	-1.347	2.458	3	7

Note. Several indicators were negatively skewed, which is consistent with generally favorable GenAI evaluations. No transformations were applied.

Appendix F. Data Screening Flow

Table F1. Data Screening Summary

Screening step	n	Decision / note
Raw recorded Qualtrics responses	83	Starting export
Preview/test records removed	3	Not real study responses

Screening step	n	Decision / note
Declined consent removed	1	Consent criterion
Ineligible respondents removed	9	No prior academic GenAI use
Failed attention check removed	7	Did not select Agree as instructed
Missing-data exclusions	0	None in retained model variables
Speeders removed	0	Diagnostic screening only; none flagged
Exact duplicate substantive patterns removed	0	None found
Final usable sample	63	Used for analysis

Note. Median completion time was 216 seconds. The one-third-median diagnostic flag was 72 seconds. The shortest retained case was 74 seconds.